

A Generalization of Rolle's Theorem: Homological Lower Bounds for number of Critical Points

(critical point of $f : M \rightarrow \mathbb{R} = df_x = 0$)

We try to understand how the topology and singularities of a prescribed zero set $N = f^{-1}(0)$ force critical points of f . Singular points of N force critical points by the implicit function theorem, while each compact component of $M \setminus N$ contributes lower bounds coming from Morse theory.

Background:

- Recall Rolle's theorem: if $f : D \rightarrow \mathbb{R}$ is C^2 where $D \subset \mathbb{R}^n$ is compact connected, and $f^{-1}(0) = \partial D$, then f has a critical point in D ($f'(x_0) = 0$). The proof is easy by considering a point which reaches maximum/minimum value.
- The homework contains such a problem: if $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is C^2 and $Z = f^{-1}(0)$ is the figure-eight graph, how many critical points must f have? Solution: The answer is 3. The center of Z is not locally Euclidean, so it must be a critical point, otherwise the implicit function theorem gives a diffeomorphism to $\mathbb{R}^1$. By Rolle's theorem, each of the bounded components of $\mathbb{R}^2 \setminus Z$ has critical points, so there are at least 3 critical points. Also, $F(x, y) = (x^2 + y^2)^2 - (x^2 - y^2)$ has exactly 3 critical points. We try to generalize this result, to arbitrary smooth manifolds M and subcomplexes N , and determine for smooth functions $f : M \rightarrow \mathbb{R}$, $f^{-1}(0) = N$, the least number of critical points of f .

For example, for $N = T^2$ be the torus in $\mathbb{R}^3$, if $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ satisfy $f^{-1}(0) = N$, then f has at least **two** critical points. This is better than Rolle's theorem, which only gives one critical point.

We only consider Morse functions here, for general smooth functions, the same homological lower bound should be interpreted with multiplicity: an isolated degenerate critical point may contribute several dimensions of local homology. Equivalently, after a small perturbation, such a point can split into several non-degenerate Morse critical points, as in the proof of Poincare-Hopf theorem. Thus the bound counts Morse critical points in the generic case, and counts critical points with multiplicity in the general case.

Step 1

Recall **Morse theory** studies smooth manifolds M using critical points of Morse functions $f : M \rightarrow \mathbb{R}$. We have the following result:

Theorem: The CW structure of M corresponds to the critical points of $f : M \rightarrow \mathbb{R}$ with non-degenerate critical points. A Morse function on a compact smooth manifold M induces a CW decomposition of M , with one λ -cell for each critical point of index λ .

We can derive the following bound using cell homology:

Theorem (weak Morse inequality): Let M be a compact smooth manifold, $f : M \rightarrow \mathbb{R}$ a Morse function, and $c_k = \#\{\text{critical points of index } k\}$. Then for each k , $c_k \geq b_k(F) = \dim_F H_k(M; F)$ for any field F .

Examples:

- For $M = \mathbb{S}^n$, this n -dimensional manifold has 0-cells and n -cells, so there are at least two critical points. The height function $f : \mathbb{S}^n \rightarrow \mathbb{R}$, $(x_0, \dots, x_n) \mapsto x_0$ has exactly **two** critical points. **Furthermore**, recall

Reeb's theorem that if $f : M \rightarrow \mathbb{R}$ is a Morse function with exactly two critical points, and M is compact, then M is homeomorphic to $\mathbb{S}^n$ (though maybe not with standard differential structure).

- For the torus $M = T$ embedded in $\mathbb{R}^3$, $\chi(M) = 0$ so the least cell structure is one 0-cell, two 1-cells and one 2-cell. Therefore there are at least **four** critical points, and the height function reaches this bound.
- For $M = \mathbb{RP}^2$, let $F = \mathbb{F}_2$ then $b_0 = b_1 = b_2 = 1$ so there are at least three critical points. $f([x : y : z]) = \frac{x^2 + 2y^2 + 3z^2}{x^2 + y^2 + z^2}$ has exactly **three** critical points.

Step 2

Now we consider adding the requirement $f^{-1}(0) = N$:

By the example of $M = \mathbb{R}^2$ and $N = \text{figure-eight}$, there are two reasons new critical points are introduced:

- N is not locally Euclidean at some point x_0 , then $f^{-1}(0) = N$ implies x_0 must be a critical point by the implicit function theorem.
- $M \setminus N$ is divided into several components, each given a boundary condition.

The first part is easy to count. For the second part, we need the relative Morse inequalities:

Theorem: Given a CW-pair (M, A) where M is a compact smooth manifold, the number of critical points of a Morse function $f : M \rightarrow [0, \infty)$ such that $f^{-1}(0) = A$ is at least $\sum_k \text{rank} H_k(M, A)$ (or $\sum_k \dim_F H_k(M, A; F)$ for a field F).

Suppose $N = f^{-1}(0)$ has finitely many singular points S , and for each compact component U of $M \setminus N$, the closure $\bar{U}$ is a compact manifold with boundary. If f is Morse on each such U , then

$$\# \text{Critical}(f) \geq |S| + \sum_U \sum_k \dim_F H_k(\bar{U}, \partial \bar{U}; F).$$

By Lefschetz duality (assuming orientability over F), $H_k(\bar{U}, \partial \bar{U}; F) \cong H^{n-k}(\bar{U}; F)$, so the number of critical points is at least

$$|S| + \sum_U \sum_k \dim_F H^k(\bar{U}; F).$$

where then sum is over components whose closures are compact manifolds with boundary.

Examples:

- Let $M = \mathbb{R}^3$ and $N = T^2$ be the embedded torus.
The standard Rolle's theorem only shows if $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $f^{-1}(0) = N$ then f has a critical point. However, the bounded component U of $M \setminus N$ is $S^1 \times D^2 \simeq S^1$, hence $H^0(\bar{U}; F) = F$ and $H^1(\bar{U}; F) = F$. Therefore we obtain a better result, that f has at least two critical points.
Equality holds for $g(\theta, u, v) = (1 - u^2 - v^2)(1 + \varepsilon \cos \theta)$, $0 < \varepsilon < 1$ which can be extended to the unbounded component.
- Let $M = \mathbb{S}^2$ and N be three disjoint latitude circles, then $M \setminus N$ consists of two disks and two annuli, so there are at least $1 + 2 + 2 + 1 = 6$ critical points.

- Let $M = T^2$ and N be m horizontal circles and n vertical circles, then there are mn points in N that are not locally smooth hypersurfaces, and $M \setminus N$ consists of mn disks, so f has at least $2mn$ critical points.